

Suppose a **Faster R-CNN** based object detector predicts 7 bounding boxes on a sample of 3 images with 5 ground truth annotations. The prediction confidence for each of the bounding boxes along with the **TP / FP** labels are given in the following table:

Images	Confidence (%)	TP / FP Label
Image 1	92	FP
Image 1	78	TP
Image 2	66	FP
Image 2	87	TP
Image 2	93	TP
Image 3	74	TP
Image 3	63	FP

For each of the predictions,

- Determine the accumulated TP, FP values. [5]
- Calculate the **Precision** values. [7]
- Calculate the **Recall** values for each of the predictions. [7]

Sorted								
Images	%	TP/FP	TP	FP	Acc TP	Acc FP	Precision	Recall
Image 2	93	TP	1	0	1	0	1/1	1/5
Image 1	92	FP	0	1	1	1	1/2	1/5
Image 2	87	TP	1	0	2	1	2/3	2/5
Image 1	78	TP	1	0	3	1	3/4	3/5
Image 3	74	TP	1	0	4	1	4/5	4/5
Image 2	66	FP	0	1	4	2	4/6	4/5
Image 3	63	FP	0	1	4	3	4/7	4/5